

Optimal Location of Electric Vehicle Charging Station in WSCC 9-Bus System Using Power Flow Analysis

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Abstract - The increasing penetration of electric vehicles (EVs) adds significant load demand to power systems, making the optimal location and sizing of EV charging stations an important planning consideration. Improper placement may lead to increased power losses and voltage deviations, thereby affecting overall system performance. This project investigates the optimal location of an EV charging station in the WSCC 9-bus test system using power flow (PF) analysis. A base case load flow study is first performed on the standard WSCC 9-bus system to establish reference operating conditions. The EV charging station is modelled as an additional load with specified active and reactive power demand. Three different cases are considered by placing the EV charging station at different load buses while keeping the transmission line data unchanged. For each case, power flow analysis is conducted to evaluate the impact on voltage profile and transmission losses. The results demonstrate that the location of the EV charging station significantly influences system voltage stability and power losses. The study identifies the most suitable bus location based on improved voltage profile and reduced losses, providing guidance for effective EV charging infrastructure planning in power systems.

Keywords – *Electric Vehicles (EVs), EV Charging Station, Optimal Location, Power Flow Analysis, WSCC 9-Bus System, Voltage Profile, Transmission Losses*

I. INTRODUCTION

The rapid growth of Electric Vehicles (EVs) is transforming the modern power system by introducing additional load demand. With increasing environmental concerns and the need to reduce carbon emissions, EV adoption is rising significantly. However, the integration of EV charging stations into existing power systems poses several technical challenges.

One of the major challenges is the optimal placement of EV charging stations. Improper placement can lead

to increased transmission losses, voltage deviations, and reduced system efficiency. Therefore, it is essential to analyse the impact of EV charging load on power systems and determine the most suitable location for its integration.

Power flow analysis plays a crucial role in evaluating the performance of a power system under different loading conditions. Among various methods, the Newton-Raphson method is widely used due to its high accuracy and fast convergence. In this project, the WSCC 9-bus test system is used as a standard benchmark network for analysis.

The EV charging station is modelled as an additional load with both active and reactive power components. Different cases are considered by placing the EV load at various buses, and their impact on voltage profile and transmission losses is analysed.

To further improve the results, Particle Swarm Optimization (PSO) is used to determine the optimal distribution of EV load among buses. The combination of power flow analysis and optimization helps in identifying the best location for EV charging stations, ensuring improved system performance and reduced losses.

II. PROPOSED METHOD

The proposed method focuses on identifying the optimal location of an EV charging station in the WSCC 9-bus system using power flow analysis and optimization techniques.

Initially, a base case load flow analysis is performed without the EV load to establish reference operating conditions. After that, the EV charging load is added to selected load buses such as Bus 5, Bus 6, and Bus 8. The system is analysed for each case while keeping the

transmission parameters constant.

The Newton-Raphson method is used to solve the power flow equations and determine bus voltages, power flows, and transmission losses. Based on these results, the impact of EV placement is evaluated.

Further, Particle Swarm Optimization (PSO) is applied to optimize the reactive power distribution of EV loads. The objective is to minimize transmission losses while maintaining voltage stability. The optimal solution is obtained by comparing results from different cases and selecting the best-performing bus.

III. METHODOLOGY USED

- (i) System Modelling Domain
- (ii) Analysis Domain
- (iii) Optimization Domain

SYSTEM MODELING DOMAIN

In this domain, the WSCC 9-bus system is modelled using given bus and transmission line data. The system includes generator buses, load buses, and transmission lines represented by impedance values.

The EV charging station is modelled as an additional load connected to specific buses. Both active and reactive power components of EV load are considered for accurate analysis.

ANALYSIS DOMAIN

In this domain, load flow analysis is performed using the Newton-Raphson method. The system is analysed under different operating conditions to evaluate:

- Bus voltage magnitude
- Voltage angle
- Power flow in transmission lines
- Transmission losses

The base case and different EV placement cases are compared to study the system behaviour.

OPTIMIZATION DOMAIN

In this domain, Particle Swarm Optimization (PSO) is used to find the optimal distribution of EV reactive power among buses.

Each particle represents a possible solution, and the algorithm iteratively updates the solution based on personal and global best values. The objective function is to minimize total transmission losses.

IV. EQUIPMENTS / SOFTWARE USED

The following tools and software are used in this project:

- MATLAB for simulation and analysis
- Power system data (bus and line data of WSCC 9-bus system)
- Newton-Raphson load flow algorithm
- Particle Swarm Optimization (PSO) algorithm

V. WORKING

The working of the proposed system is based on a systematic evaluation of the power system under different operating conditions with and without the integration of Electric Vehicle (EV) charging load. Initially, the standard WSCC 9-bus system is modelled by inputting the required bus data and transmission line data into the simulation environment. These data include parameters such as bus type, voltage magnitude, active and reactive power demand, line impedance, and line charging values. Based on this data, the Y-bus matrix is formed, which represents the admittance of the entire network and serves as the foundation for power flow analysis.

After forming the Y-bus matrix, a base case load flow analysis is performed using the Newton-Raphson method to determine the initial operating conditions of the system. This includes calculating bus voltages, voltage angles, power flows, and transmission losses without the presence of EV load. The base case results act as a reference for comparing system performance under different EV integration scenarios.

Subsequently, the EV charging station is modelled as an additional load and is introduced at selected load buses, specifically Bus 5, Bus 6, and Bus 8. For each placement, the load flow analysis is repeated using the Newton-Raphson method to evaluate the impact on system parameters such as voltage profile and transmission losses. Since EV charging introduces both active and reactive power demand, its effect on system stability becomes significant and must be carefully analysed.

To further enhance system performance and determine the most optimal configuration, Particle Swarm Optimization (PSO) is employed. PSO is used to optimally distribute the reactive power component of the EV load among the selected buses in such a way that total transmission losses are minimized while maintaining acceptable voltage levels. The optimization process iteratively searches for the best solution by updating particle positions based on individual and global best values.

Finally, the results obtained from different cases are compared in terms of voltage stability and transmission losses. Based on this comparative analysis, the optimal bus location for EV charging station placement is identified. The system continuously evaluates performance using these criteria, ensuring efficient and reliable operation of the power network.

VI. ALGORITHM

The algorithm used in this project combines the Newton-Raphson load flow method with Particle Swarm Optimization (PSO) to determine the optimal placement of the EV charging station. Initially, bus data and line data of the WSCC 9-bus system are provided to form the Y-bus matrix, which represents the network admittance.

Voltage magnitudes and phase angles are initialized, and the mismatch between scheduled and calculated active and reactive power is computed. Using these mismatches, the Jacobian matrix is formed, and corrections are applied iteratively until the error falls within the specified tolerance, ensuring convergence of the load flow solution.

After obtaining the base case results, the EV charging load is introduced into the system. The PSO algorithm is then applied to optimize the reactive power distribution among selected buses. Each particle represents a possible solution, and its fitness is evaluated based on minimizing transmission losses.

Particles update their positions using personal best and global best values, allowing the algorithm to converge toward the optimal solution. Finally, the optimized results are used to compute voltage profile and transmission losses, ensuring improved system performance and stability.

The working algorithm is shown in figure below:

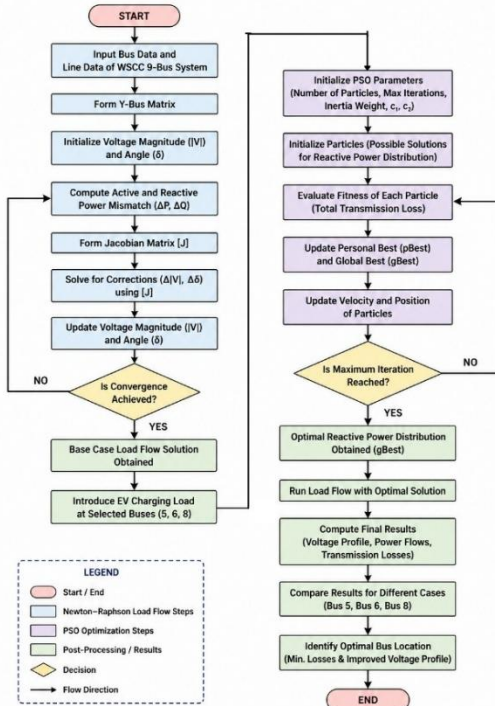


Fig 1: Newton Raphson Load Flow with PSO optimization

VII. CONDITIONS OF WORKING ALGORITHM

The working of the proposed algorithm is governed by a set of predefined conditions to ensure accurate analysis and proper evaluation of system performance under different scenarios. These conditions are necessary to maintain consistency in results and to allow fair comparison between different EV charging placement cases.

The main conditions considered in this project are as follows:

1. **EV Load Placement Condition:** The EV charging load is added only to selected load buses, namely Bus 5, Bus 6, and Bus 8. These buses are chosen based on their load characteristics and suitability for EV integration.
2. **Constant Load Condition:** The total EV load (active and reactive power) is kept constant for all case studies. This ensures that the variation in results is due only to the change in location and not due to change in load magnitude.
3. **Transmission System Condition:** The transmission line parameters such as resistance, reactance, and line charging are kept unchanged for all cases. This helps in isolating the effect of EV load placement on system performance.
4. **Load Flow Convergence Condition:** The Newton-Raphson method is used for power flow analysis, and convergence is achieved when the mismatch error is reduced below a specified tolerance limit.
5. **Optimization Constraint Condition:** In PSO, constraints are applied to ensure that the total reactive power of EV load is within permissible limits. The sum of reactive power distributed among buses should not exceed the defined maximum value.
6. **Performance Evaluation Condition:** The system performance is evaluated based on voltage profile and total transmission losses. Acceptable voltage limits are maintained to ensure system stability.
7. **Optimal Solution Condition:** The optimal bus location is selected based on minimum transmission losses and improved voltage profile obtained after applying the PSO optimization.

VIII. CONCLUSION

The project analyses the impact of EV charging station placement on power system performance using the WSCC 9-bus system. The results show that improper placement of EV load leads to increased transmission losses and voltage deviations, affecting overall system efficiency. The Newton-Raphson method effectively solves the power flow problem, while Particle Swarm Optimization (PSO) helps in identifying the optimal solution.

Based on the analysis of different cases, the optimal bus location is determined by considering improved voltage profile and reduced transmission losses. This ensures better system stability and more efficient operation of the power network.

IX. FUTURE SCOPE

There is always scope for improvement in power system planning, especially with the rapid growth of Electric Vehicles (EVs). The proposed work can be further extended in several directions to enhance system performance, efficiency, and real-world applicability. Some of the future scope areas are as follows:

- Integration of multiple EV charging stations at different buses to study their combined impact on voltage profile and system stability.
- Optimal sizing of EV charging loads along with location determination to achieve better load management and reduced transmission losses.
- Integration of renewable energy sources such as solar and wind along with EV charging infrastructure to promote sustainable and green energy systems.
- Analysis of larger and more complex IEEE bus systems (such as 14-bus, 30-bus, or 118-bus systems) for more realistic and scalable results.
- Implementation of real-time monitoring and control using smart grid technologies to improve system reliability and operational efficiency.
- Application of advanced optimization techniques such as Artificial Intelligence (AI) and Machine Learning (ML) for more accurate prediction and optimization of EV charging demand.

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